# Tutorial 2 (to be done in week2)
## Data Modeling (Part 2) & Relational Set Operators - Guide

### Tutorial Objectives:

- Transform conceptual model into a logical model using Visio
- Learn about relational set operators

### Tutorial Tasks:

a. **Arrive at a logical model by mapping the conceptual model (Tiny College example) to the chosen relational constructs (tables, primary and foreign keys, etc.).**
b. **Apply relational set operators in Oracle 12c to a given problem domain. Initial SQL script file is provided.**
c. **Analyse some relational operators using the script provided.**

These tutorial tasks can be attempted reasonably well by following the given instructions. Make sure you pay attention to the "Notes" which have hints and/or some explanation. Some code learnt in the lectures can be easily adapted. Specific lecture slides are also referred for Part B. Some of the Part B questions may be slightly difficult here in terms of the SQL (for many of you), make sure you revisit these after learning some more coding. They are given here because the concept is part of the corresponding lecture (union, joins, etc.). We will be learning SQL in the coming weeks so that should help. It is always a good idea to incrementally test your code. Do one question at a time (may run a statement at a time rather than all the code). Initially it also helps to fix a mistake and then run, rather than trying to fix all mistakes as at times there is a cascading affect while debugging.

Remember you can also discuss any of the tutorial elements via forums when stuck. Would be better to provide your context so that it is easier to explain/debug (e.g., relevant screenshots are useful, some code – as then it can be copied and pasted).

a. Is asking you to think about the PKs (should always look at the data for better understanding)
b. Go and make changes to the code. Add the PK constraint and see the affect. Can you now add two rows with same PK?
c. Two decimal places – try to violate (e.g., add a number bigger than the permitted constraint) while inserting as well and see what is being said in return (important to learn to decipher to debug).
d. Again, see the affect by violating.
e. The information will not make much sense, but question is given to understand the Union concepts (compatible, etc).
f. Bit challenging, many ways to attempt the code as you will notice in coming weeks with SQL.
g. Again – somewhat reverse of the above question. But one should be able to figure out the expected answer even if the right code cannot be written (with

ease) at this stage. Once again, the concepts are important, and code can be revisited a bit later.

For Part C questions you can (**later)** run the code to confirm your answers. It is important to understand concepts (union compatible, union, union all, intersect, minus, etc.) Order in which operations are carried out (e.g., use of parenthesis, etc) is also important part of the process (how to arrive at an answer using a dry run).